EXATHON INDIA 2026 · TRACK 3

VoltProof

Traceable evidence of power cuts and poor supply, from minute-by-minute voltage records, analysed inside Exasol.

> Households and small shops on unreliable electricity can't show when, and for how long, their power failed, because when they complain, their own memory is usually all they have to set against the utility's records. If this worked, they could take a dated, minute-by-minute record of every outage and voltage problem, with its uncertainty stated, to their utility or grievance forum.

Data: Prayas (Energy Group), Electricity Supply Monitoring Initiative (ESMI), Harvard Dataverse doi:10.7910/DVN/CLLZZM (CC0 1.0).

github.com/stredevirad/voltproof-exathon-2026 · demo video: stredevirad.github.io/voltproof-exathon-2026

THE PROBLEM

Evening power cuts, and nothing to show for them

Illustrative

“The power goes off almost every evening.”

A shopkeeper's memory, against the utility's own records.

A shopkeeper in a small Uttar Pradesh town loses business to evening cuts. When he complains, he cannot show **when** the power failed, **for how long**, or **whether his neighbours lost it too**.

Daily cuts and voltage too low to use are common in parts of India, and complaints rarely have an independent, dated record behind them.

WHY THIS MATTERS

Many consumers, and a possible future source of readings

2.74 crore

domestic electricity consumers in Uttar Pradesh alone, as the then UPPCL chairman said on 27 January 2023

We do not claim that every one of them has outages or would use VoltProof. Most households have no monitor like ESMI's, so today VoltProof is a method, not a service.

7.24 crore

smart meters (consumer, distribution transformer and feeder meters) installed in India as of 30 June 2026; 20.33 crore sanctioned under the Revamped Distribution Sector Scheme

A possible future data source for the same method. VoltProof does not use, or have access to, smart-meter data today.

Sources. Left: Drishti IAS, 28 January 2023 (drishtiias.com, “Uttar Pradesh became the state with maximum electricity connections in the country”). Right: a written reply in Parliament by the Minister of State for Power, reported by T&D India, 6 August 2026 (tndindia.com). Links in the README.

THE RECORD ALREADY EXISTS

Independent monitors recorded the voltage every minute

- Prayas (Energy Group)'s ESMI monitors were plugged in at homes and shops and recorded the voltage every minute. VoltProof uses Uttar Pradesh, January 2016 to June 2019.
- The readings are published openly, but as raw hourly files with gaps, duplicates and defects: hard to use, and impossible to show to a utility as they are.

102.8 million

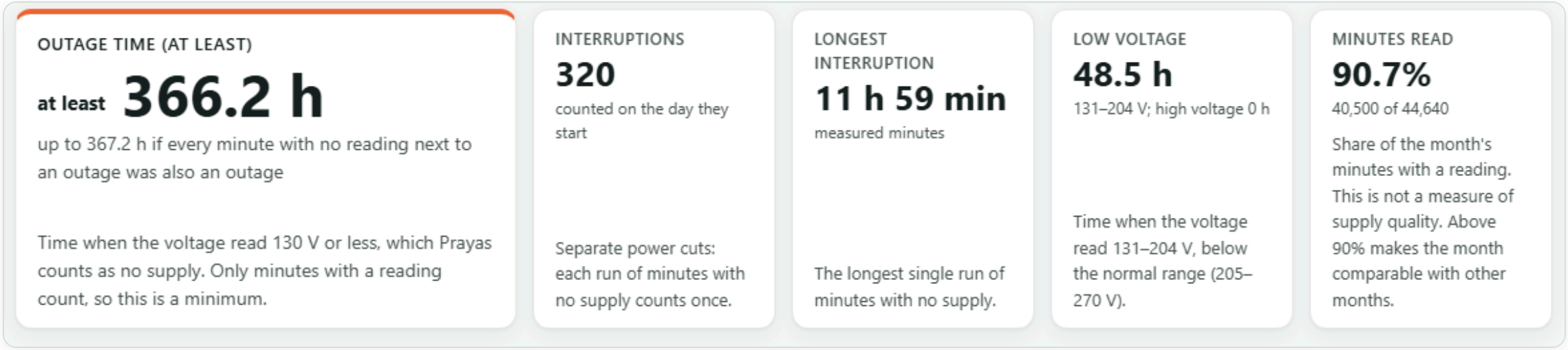
minutes of clean readings from Uttar Pradesh that VoltProof analyses

`Location name, Date, Hour, Min 0, Min 1, … Min 59`

the published files: one row per hour, holding that hour's 60 minutes

WHAT VOLTPROOF GIVES HIM

Every cut, dated to the minute, with its certainty stated



- One monitor, one month: **at least 366.2 hours** without supply, every cut listed.
- Missing data is never counted, so the headline is a minimum.
- Open any day to see each minute; open any cut to see its source lines.
- One click: a PDF evidence report to take to the utility.
- Anyone can analyse readings from their own monitor today, the same way; the app labels them “not verified”.

DEMO MOMENT · BAHRAICH, 25 JULY 2018, 18:27

One evening in Bahraich: 17 monitors lost supply within 2 minutes

16 other monitors lost supply within 2 minutes

Potential shared supply event Bahraich | 2018-07-25 18:27: 136 coincident pairs in Bahraich district. 21 monitors in the district had interruptions that month.

Monitor	Start	Offset	Measured
Huzurpur-Bahraich	**2018-07-25 18:27**	**+0 min**	**1 h 07 min**
Mihinpurwa-Bahraich	2018-07-25 18:27	+0 min	37 min
Agarwal Mohalla Jarwal Kasba	2018-07-25 18:29	+2 min	32 min
Bahadurpur-Bahraich	2018-07-25 18:29	+2 min	1 h 10 min
Bardaha-Bahraich	2018-07-25 18:29	+2 min	49 min

Chance comparison, Bahraich 2018-07 (comparable months only): **3,945** coincident pairs observed, about **1827.5** expected if the monitors' outage days were shifted against each other by whole days (time of day kept; 20 fixed shifts); none of 20 shifted variants reached it.

- Huzurpur-Bahraich plus 16 other Bahraich monitors lost supply between 18:27 and 18:29 (the panel lists the 16 others).
- VoltProof calls this a **potential shared supply event**: the cuts started together. It names no cause and no equipment.
- Chosen by a fixed, documented rule, not hand-picked: the evening potential shared supply event with the most monitors in June–August 2018.

The whole month, not just this evening: in Bahraich in July 2018, interruptions at two different monitors started within 2 minutes of each other 3,945 times; about 1,827.5 would be expected by chance (20 versions with the days shifted; none reached the observed count).

Coincidence is not cause.

HOW IT WORKS, IN PLAIN WORDS

Four steps, from published files to a report

1

Check

Every file's checksum (a fingerprint of its contents) must match the published one before it is used.

2

Load

The raw rows go into the database unchanged, each one keeping its file and line number.

3

Analyse

Every minute is sorted into no supply, low, normal, high or no reading; each cut is found, measured and compared with other monitors.

4

Show

The app and the PDF only read the results. Every number points back to its file line.

Steps 2 and 3 run inside Exasol (next slide).

Missing data is never an outage. A cut next to a gap gets a measured minimum (the headline) and a possible maximum.

The voltage classes are Prayas's published ones: no supply at 130 V or less, low 131–204 V, high above 270 V. A month is compared only when more than 90% of its minutes were read.

HOW WE USE EXASOL PERSONAL

The whole analysis runs inside Exasol, on a laptop

EXASOL FEATURE	WHAT WE DO	MEASURED OR TESTED
Native IMPORT	Loads the raw 63-column rows unchanged, each with its file id and line number	7,367,581 raw rows in 30.3 s
Unpivot with CONNECT BY	A 60-row generator turns each hourly row into 60 minute rows	102,927,720 minute rows in 11.0 s
One window function	Finds every run of one voltage class with a single ROW_NUMBER window	665 s before our rewrite, 35.6 s after
Bucketed equality joins	Pairs interruptions in a district that start within 2 minutes, as hash joins on a time bucket	184,299 pairs in 0.4 s; chance baseline in 3.9 s
Audit log	Matches every recorded statement by its exact SQL text; shows each one's time and memory peak, measured by Exasol, in the app and the PDF	0 different, 0 missing in every benchmark build
Users and schemas	Each upload in its own schema under a restricted user; a read-only user sees only seven commented views	Tests: neither can read anything else
exasol/nano in CI	Every push starts Exasol's own image, builds the SYNTHETIC set with the same SQL and checks it against the oracle	Fixtures, synthetic set, provenance, browser smoke test

210.4 s for the full build of the Uttar Pradesh data (39 SQL statements) on a 15.6 GB laptop. Python only checks files, starts the SQL steps and shows results.

ASK VOLTPROOF

An AI client can ask the results, read-only

AN AI CLIENT
asks a question in plain words

→

EXASOL'S MCP SERVER
open source; read queries only

→

VP_READER
a database user that may only read

→

SEVEN VIEWS
months, events, shared events, days, quality, provenance, rules

- Every view and column carries a comment, so the client knows what each number means.
- It cannot read a table, an upload or another schema, and cannot write: tested.
- **The agent's sentences are its own.** VoltProof claims only what the views show; the app and the reports contain no generated text.

Asked through the server, 29 Oct: which monitors lost supply with Huzurpur-Bahraich on the evening of 25 July 2018?

Answer from the view: the group **Bahraich | 2018-07-25 18:27**, 17 monitors.

A DELETE was refused by the server; reading a table was refused by the database.

WHY YOU CAN TRUST THE NUMBERS

Every number can be checked, not just believed

A NUMBER
in the app or the PDF report

→

THE SQL STEP
that computed it, and the hash of its SQL file

→

THE EXASOL STATEMENT
recorded in the audit log

→

THE SOURCE LINE
in the published file

- A separate Python check, written independently from the same rules, gives identical results on every test. It is a test, not the product.
- Hand-written test cases, random test data and real slices of the data are all checked the same way.
- Results are compared with Prayas's own published figures, including where they differ (slide 12).

HONEST ABOUT THE DATA

Defects in the published files: found and labelled, never hidden

2019 January–June ! lower bounds

Hours that read 0 V throughout were left out of this file. The gaps stay missing data, so 2019 figures are labelled lower bounds in the app and the reports.

2019 July–December: Hour column lost

Almost every row says hour 0, so the minute of a reading cannot be known. The file is loaded, checked and excluded, with that reason.

2019 names and dates

Some locations gained an added " [Offline]"; they are matched to their earlier names only when that is unambiguous. Dates changed format between files and are read per file.

Found by checking the data against Prayas's own single-monitor charts. Details and counts: appendix A4.

CHECKED AGAINST PRAYAS'S PUBLISHED FIGURES

Single monitors agree; our daily averages are higher

2.0–2.6×

our daily-outage averages, compared with Prayas's Figure 2

We measure more outage than Prayas's Figure 2, and say so. Single-monitor profiles agree within 0.2 h/day, and so do the evening and time-of-day charts.

Our daily-outage averages are higher than Prayas's Figure 2. Prayas's Figure 2 and Figures 8–9 imply different averages, and the article does not state the averaging used for Figure 2. No parameter was tuned to close the gap.

The full comparison table is in appendix A3.

LIMITS AND LEGAL FRAMING

What VoltProof does not claim

- A plug-in monitor is not the utility's meter; an outage may also come from a tripped breaker, wiring or the monitor.
- ESMI data is historical (the whole dataset: 2014–2019; used here: Uttar Pradesh, January 2016 to June 2019): it describes supply then, not today.
- A potential shared supply event is a coincidence in time, not proof of a common cause.
- No amount owed and no breach is stated. VoltProof is an evidence-analysis aid, not legal advice.

Grievance route the report points to

In Uttar Pradesh, the Commission's 2003 regulations provide a Consumer Grievance Redressal Forum and an Electricity Ombudsman.

UPERC CGRF and Ombudsman Regulations, 2003; PIB, Ministry of Power, 30 Dec 2010.

WHO COULD USE IT NEXT

Who could use it, and what comes next

Households and small shops

A dated record of every cut, with its certainty, instead of a memory, to take to their utility or grievance forum. Anyone can also upload readings from their own monitor today; the app labels them “not verified”.

Consumer groups, researchers, journalists

Event-level evidence and a report per location-month, instead of raw minute files. Today that means the ESMI records.

A grievance forum or a regulator

Each number traces to a file line and an Exasol statement, so it can be checked rather than trusted.

Next (not claimed today): New monitors or smart-meter data through the same SQL; each state's standards of performance, from the primary regulation, to show which events may warrant review through the grievance forum.

More states and maps: the ESMI files cover many states; a map needs coordinates the source does not have. For a household without a monitor, VoltProof is a method today, not a service.

In one line: 102.8 million minutes of independent readings, analysed inside Exasol, every number traced to its SQL step and source line.

TEAM

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With thanks to Prayas (Energy Group) and the ESMI volunteers who collected and published these readings.

102.8 million minutes, full build in 210.4 s

30.3 s
native IMPORT of 7,367,581 raw rows (11 files)

11.0 s
in-database unpivot to 102,927,720 minute rows

35.6 s
gaps-and-islands over 102.8 million readings (one window function)

SCALE FIX (RECORDED IN EXASOL'S AUDIT LOG)	BEFORE	AFTER
Run detection: LAG + running SUM → one ROW_NUMBER window	665 s, 7.4 GiB temp	35.6 s
Duplicates: minute-level GROUP BY → hour-level first	1,892 s, then a cancelled step	13.5 s
Chance baseline: band join → bucketed equality joins	348 s (sample only)	3.9 s (full data)

In all 6 benchmark builds, every statement's session and statement id matched EXA_DBA_AUDIT_SQL by exact SQL text (0 different, 0 missing); the same audit statistics produce these benchmarks. Sources: docs/BENCHMARKS.md, docs/WHY_EXASOL.md.

The SQL must equal an independent implementation, exactly

- **18 hand-written fixtures** (rules R-1 to R-21), expected values derived by hand.
- **Independent plain-Python oracle**: identical to the SQL on every fixture, on **40 random Hypothesis batches**, and on real Uttar Pradesh slices (4,930 events; 20,127 coincident pairs).
- **Harness sensitivity**: a single wrong value is caught.
- **Report consistency**: every number in the PDF equals an independent SQL query and the rendered page.
- **Provenance**: every statement of every build matched to Exasol's audit log by exact SQL text (0 different, 0 missing in all 6 benchmark builds).
- **End-to-end** browser tests of the guided demo, plus a scan for blocked wording on every page and report.

docs/VALIDATION.md §1 · docs/TEST_REPORT.md

APPENDIX A3 · VALIDATION AGAINST PRAYAS'S PUBLISHED FIGURES

Agrees at the level of single monitors; headline averages differ

PRAYAS FIGURE	PRAYAS	VOLTPROOF	
Single-monitor profiles, 2017–2018 (4 monitors, 34 monitor-months)	hours/day per class	largest difference 0.2 h/day	✓ agrees
Evening outage (Figure 9), min: megacity / other municipal / rural	12 / 53 / 144	11.3 / 47.8 / 147.3	✓ agrees
Outage by time of day (Figure 8), summed, min/day	≈ 56 / ≈ 155 / ≈ 523	56 / 174 / 555	✓ agrees
Daily outage (Figure 2), min/day	21 / 87 / 260	55.6 / 174.4 / 555.2 2016–2018 only: 56.3 / 189.5 / 566.3	≠ higher

Our daily-outage averages are higher than Prayas's Figure 2; Prayas's Figure 2 and Figures 8–9 imply different averages, and the article does not state the averaging used for Figure 2. No parameter was tuned to close the gap.

docs/VALIDATION.md §3. Primary definition, fixed before comparing: comparable months, 2016-01 to 2019-06, pooled. Headline figures elsewhere use 2016–2018 (the 2019 file defect, D-031); that variant was already reported in §3.3.

Found, documented and handled, never silently dropped

2019 Jan–June: all-0 V hours omitted ! lower bounds

In 2016–2018, 11–17% of read location-hours are entirely 0 V; in 2019, 4.3%. 82% of 2019 gaps have 0 V just before and after. Found by comparing with Prayas's own single-monitor charts.

Handled: gaps stay missing data; 2019 figures are labelled lower bounds in the app and reports; headlines use 2016–2018.

2019 July–Dec: Hour column lost

690,915 of 695,974 rows say hour 0, so minute timestamps cannot be determined. Loaded, verified, excluded with that reason.

2019 formats

Dates M/D/YYYY (earlier DD-MM-YYYY), detected per file; names with an added " [Offline]", matched only when unambiguous.

Measured minutes only, so these are lower bounds

Rural Uttar Pradesh, 2016–2018

566.3 min/day

measured outage (lower bound), 82 rural monitors, comparable months, VoltProof

Prayas's own article reports "in excess of 4 hours (>240 minutes) a day" for rural areas; our measurement is higher (see validation).

Sources: docs/HEADLINE_FIGURES.md; Prayas, "ESMI in Uttar Pradesh", 7 Aug 2019.

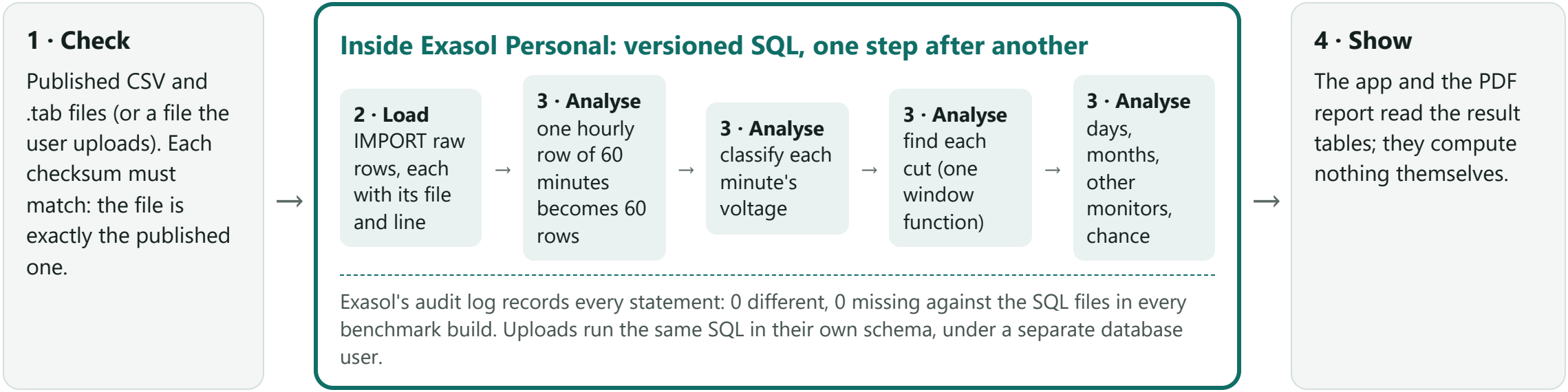
Potential shared supply events

Across the 276 district-months with at least two monitors, interruptions started within 2 minutes of each other 91,532 times; the time-shifted baseline expects 40,704.6.

150 of those district-months are above all 20 shifted variants; by pure chance, at most about 1 in 21 would be (about 13).

Coincidence is not cause.

The whole analysis runs inside Exasol, on a laptop



210.4 s

full build of the Uttar Pradesh data (39 SQL statements), on a 15.6 GB laptop

In Exasol: 102,927,720 minute rows built in 11.0 s; every cut found in 35.6 s (665 s before our own fix, both timed from Exasol's audit log).

The published files hold all states; all are loaded, and only the Uttar Pradesh rows become minutes. Python only checks files, starts the SQL steps and shows results. More timings: appendix A1.